

Incentivizing AI Evaluation through Insurance: A Mechanism Design Approach

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Abstract

Insurance is an attractive instrument for AI governance, as premiums price risk into deployment decisions and coverage compensates harmed parties without continuous regulatory oversight. Such proposals stall on a pricing problem: safety evaluations cannot supply the reliable risk estimates underwriting requires. This paper illustrates how an insurance scheme can operate despite such prediction errors, adapting the fair-bet mechanism of [Zhao and Ermon, 2021], under which a lab backs its evaluation-based forecasts with side payments. Our contribution is to make evaluation a costly choice inside the model. Effort invested into evaluation determines forecast error, and the mechanism rewards sharper forecasts through a fee on the surplus the scheme creates. We find that under the scheme, optimal evaluation effort exists, is strictly positive when sharper forecasts are valuable to deployers, and rises with deployment scale. Above a threshold scale, labs strictly prefer to operate the scheme, removing the need for regulatory intervention, and deployers are weakly better off. The evaluation incentive is financed by the lab’s share of deployer surplus, however, so competition among labs weakens the very incentive the mechanism creates.

1 Introduction

As AI systems are increasingly deployed into consequential settings, such as clinical workflows, hiring pipelines, and financial infrastructure, the question of who bears the costs of failure has moved to the center of the governance debate. Insurance is an attractive answer in principle: it compensates harmed parties without requiring continuous regulatory oversight or case-by-case litigation, and by pricing premiums to risk, it makes deploying a potentially dangerous model directly costly [Szpruch et al., 2025, Trout, 2025].

In practice, proposals for AI insurance have stalled on a pricing problem. Insurance requires estimates of the frequency and severity of harm, and pre-deployment safety evaluation is the natural source of such estimates for AI systems. However, evaluations are unreliable in predicting both model capabilities and behaviors [Bengio et al., 2026], and fall far short of what underwriting requires. Evaluation results are best understood as unreliable bounds on capability. Even for anticipated harms [Bova et al., 2024], a model can harbor harmful behaviors that survive safety training while evaluations report it safe [Hubinger et al., 2024]. Post-deployment, users can design elicitation techniques to defeat safeguards that previously held [Shen et al., 2024], to the point where adversaries can compose individually “safe” models to perform dangerous tasks no single-model evaluation would catch [Jones et al., 2024]. The productive question for insurance design is therefore not how to make evaluations reliable enough to price on, but whether a compensation scheme can remain sustainable when they are not.

A mechanism-design result of Zhao and Ermon [2021] suggests that incorrect forecasts of risk need not be fatal. In their fair-bet insurance scheme, a forecaster posts a risk estimate together with an uncertainty interval and accepts any side bet consistent with that announcement; each decision-maker can then choose a bet under which her realized expected utility equals her forecasted expected utility (in other words, she may act as if the forecast were individually correct even when it is not) while an online-learning strategy guarantees the forecaster sustains no loss in the long run, even against adversarial counterparties. This project adapts that mechanism to AI deployment: the lab acts as forecaster-insurer, deployers as decision-makers, and the forecast and uncertainty interval are supplied by the lab’s evaluation. The intended contribution is twofold. First, the project introduces the costs of investing effort into sharper evaluations into the model and derives under what conditions the mechanism can motivate more rigorous evaluations. Second, we examine whether the insurance scheme is rational to operate for all parties, and observe how AI market structure may shape the mechanism’s effects.

2 Related Work

Forecasting Games and Calibration: In forecasting games, a forecast can be truthful yet incorrect, so the mechanism must have a way to correct for honest errors. Statistical calibration guarantees that forecasts match realized frequencies in the long-run average, even against adversarial outcomes [Foster and Vohra, 1998], but no forecaster can guarantee every individual forecast is accurate [Zhao et al., 2020]. The accuracy of forecasts can be evaluated through betting: a skeptic who doubts the forecast can wager against them, and the outcome of her bets measures the forecast error [Shafer, 2021]. This paper adapts this line of literature, using bets as payments to correct for incorrect forecasts so that decision-makers can act on forecasts they cannot verify [Zhao and Ermon, 2021].

Classical Insurance Theory: Economic literature surrounding insurance concerns informational problems on the *insured* side, such as adverse selection, where insurance attracts risky policyholders [Rothschild and Stiglitz, 1976], and moral hazard, where coverage itself encourages risk-taking [Arrow, 1963, Shavell, 1979]. Moreover, estimable and bounded losses are important for insurance [Berliner, 1985], as ambiguity leads to higher premiums that reflect uncertainty in pricing [Kunreuther et al., 1993]. Our setting inverts the information asymmetry problem, as the underwriter has private information about its evaluations that the insured cannot verify. Moreover, instead of dealing with exogenous uncertainty, we model evaluation effort as sharpening forecasts and derive how much effort a lab underwriting its own models would invest.

AI Insurance Policy: The advantages of insurance for AI governance is twofold: premiums price risk into deployment decisions, and insurers act as private regulators by monitoring safety for underwriting [Lior, 2021, Trout, 2025]. However, lessons from cyber insurance suggest insurers struggle to price system-specific technological risk due to a lack of loss data and technological knowledge gaps [Schwarcz and Wolff, 2025]. Given this, potential proposals for AI risk insurance accept that actuarial data cannot keep pace with the advancement of AI and instead proposes pricing on forward-looking assessments, such as performance evaluations and red-teaming results [Trout et al., 2026]. That proposal leaves open both what happens when the evaluations are wrong and why a lab would invest in making them better. We propose fair-bet insurance as a mechanism to correct for error and incentivize investment into precise forecasts.

Technical Limitations of Safety Evaluations: The field’s consensus International AI Safety Report documents significant gaps between evaluations’ predictions of model capabilities and behaviors and what deployment reveals [Bengio et al., 2026]. Some of this error can be reduced through increasing investment into evaluation; broader elicitation, increased red-teaming, and better adversarial testing can sharpen forecasts of risk [Bova et al., 2024]. However, some risks cannot be reliably measured: models with harmful traits can learn to conceal them when evaluated [Hubinger et al., 2024], users are able to elicit harmful behavior despite safeguards [Shen et al., 2024], and emergent failures can arise from the interaction of many deployed agents [Hammond et al., 2025]. We model both types of evaluation error: effort can reduce the first, while the existence of risk that survives any evaluation motivates why the insurance mechanism must compensate for error rather than try to predict it.

3 Model

We borrow the framework and notation of Zhao and Ermon [2021] and instantiate the fair-bet mechanism in the context of the deployment of AI systems, with a frontier AI lab as the forecaster, the deployers of the model as the decision-making agents, and the forecast provided by the lab’s safety evaluation.

3.1 Environment

A lab has trained a model and licenses it to a sequence of downstream deployers $t = 1, \dots, T$ (a customer-service platform, an ed-tech tutoring product, or a content-drafting tool, for instance). To concretize the model, we consider the customer-service platform integrating the model into its support product, with a single verifiable harm class of toxic output. The incident indicator $y_t \in \{0, 1\}$ records whether the harm materializes during deployment engagement t , verifiable through flagged transcripts, moderation audits, or user complaints. The true incident probability $\mu_t^* \in [0, 1]$ is chosen by Nature, as it depends on factors no pre-deployment evaluation fully observes, such as the deployer’s user population, prompt distribution, and integration choices. This true probability is never revealed; only the realized outcome y_t is. Moreover, each deployment pairs a model with a distinct deployer, user population, prompt distribution, and usage context, so no two engagements share the same μ_t^* .

We restrict attention to a binary outcome per harm class rather than adopting the categorical multiclass protocol of Zhao and Ermon [2021]. The categorical formulation requires outcome classes to be mutually exclusive, whereas distinct AI harms routinely co-occur within a single engagement. We moreover observe that a taxonomy of K harm classes can be modeled by running K copies of the binary mechanism in parallel, since payments are additive and the forecaster’s guarantee applies class by class.

Deployer t is characterized by a private deployment value $v_t > 0$ and incident damage $D_t > 0$ (brand damage, remediation cost), giving the loss function

$$l_t(a, y) = \begin{cases} -v_t + D_t y, & a = 1 \text{ (deploy)}, \\ 0, & a = 0 \text{ (do not deploy)}, \end{cases} \quad (1)$$

so the deployer’s action space is $\mathcal{A} = \{0, 1\}$, deploy or do not deploy, and losses are bounded by magnitude $M \geq \max_t \{v_t, D_t\}$. We consider $D_t \leq M$ throughout: harms exceeding the maximum insurable exposure are outside the mechanism. For instance, the customer service platform integrating the model into its support product is worth v_t to it, and a toxicity incident costs it $D_t = \$100\text{K}$ in brand damage and remediation.

3.2 The Forecasting Game and the Limits of Evaluation

At each step t , Nature reveals the deployment context $x_t \in \mathcal{X} \subset \mathbb{R}^d$ to the lab and privately selects μ_t^* with $\Pr[y_t = 1] = \mu_t^*$. The lab announces a probability μ_t and an uncertainty c_t , with $(\mu_t - c_t, \mu_t + c_t) \subset (0, 1)$. Deployer t , holding her private loss function (1), selects an action $a_t \in \mathcal{A}$. Nature then draws $y_t \sim \text{Bernoulli}(\mu_t^*)$ and the deployer incurs $l_t(a_t, y_t)$. For the customer service platform, suppose the lab’s evaluation yields $(\mu_t, c_t) = (0.30, 0.05)$, or a 30 percent chance that a toxicity incident occurs somewhere in the deployment, give or take five points. However, this forecast may be incorrect, as the platform’s users may elicit behavior the red team never sampled, and the true rate may be 0.60.

Given the information revealed by the lab’s forecast, the deployer’s expected loss under the announced probability can be written as

$$L_t^{\text{avg}} = \mathbb{E}_{Y \sim \mu_t}[l_t(a_t, Y)],$$

with $L_t^{\min}, L_t^{\max}$ representing the extremes of $\mathbb{E}_{Y \sim \tilde{\mu}}[l_t(a_t, Y)]$ over $\tilde{\mu} \in [\mu_t \pm c_t]$. The deployer would like assurance that her true expected loss, written $L_t^* = \mathbb{E}_{Y \sim \mu_t^*}[l_t(a_t, Y)]$, lies in $[L_t^{\min}, L_t^{\max}]$, so that she may act as if the individual forecast is correct, even in instances of error. Proposition 1 of [Zhao and Ermon \[2021\]](#) shows this holds for all loss functions if and only if the forecast is *individually correct*: in other words, $\mu_t^* \in [\mu_t - c_t, \mu_t + c_t]$.

However, individual correctness is provably out of reach for deterministic forecasts [[Zhao et al., 2020](#), [Barber et al., 2020](#)]. It can be recovered only by randomizing the forecast, at which point the announcement no longer can act as the probability a decision-maker should act on [[Zhao et al., 2020](#)]. Informally, estimating μ_t^* from data requires many draws of the *same* μ_t^* , and when each context is observed only once, the probability that generated it cannot be inferred from a single realization without unverifiable assumptions. In the context of AI safety violations, each deployment pairs a model with a distinct deployer, user population, prompt distribution, and usage context. Thus, no two engagements share the same μ_t^* , and the limits of pre-deployment evaluations mean that a mechanism that functions without individual calibration is necessary.

3.3 The Fair-Bet Insurance Mechanism

[Zhao and Ermon \[2021\]](#) uses prices to correct for this calibration problem. Rather than making forecasts individually correct, the lab backs each announcement with side payments such that every deployer can act as if it were. By Lemma 1 of [Zhao and Ermon \[2021\]](#), announcing (μ_t, c_t) is equivalent to a willingness to accept any bet b_t paying the deployer $b_t(y_t - \mu_t) - |b_t|c_t$ for $b_t \in [-M, M]$, as these are exactly the payments with non-positive expectation under every probability in $[\mu_t \pm c_t]$. The stage game that implements these bets is their Protocol 2, where the deployer’s guarantee is:

Proposition 1 ([Zhao and Ermon, 2021](#), Prop. 2). *If the deployer selects stake $b_t = l_t(a_t, 1) - l_t(a_t, 0)$, then her expected loss net of the side payment satisfies $L_t^{\text{pay}} \in [L_t^{\min}, L_t^{\max}]$ for any μ_t^* ; when $l_t(a_t, 1) \geq l_t(a_t, 0)$ the guarantee holds with equality at the upper end, $L_t^{\text{pay}} = L_t^{\max} = L_t^{\text{avg}} + c_t(l_t(a_t, 1) - l_t(a_t, 0))$.*

Consider again the example of the customer service platform. Suppose $\mu_t^* = 0.60$. The platform selects a stake of 100K, equal to her potential loss. Her premium is $0.35 \times 100 = 35\text{K}$, including the 5 percent uncertainty. If an incident occurs, the platform loses \$100K and collects \$100K as an insurance payout. If an incident does not occur, its only loss is paying the premium. Importantly,

the platform has a guarantee that its payoff is -\$35K regardless of whether the harm happens or if the forecast was correct.

The choice of b_t depends only on quantities the deployer knows. Because an incident of model toxicity is never beneficial to the deployer ($l_t(a, 1) \geq l_t(a, 0)$ for all a), we are in the monotone-loss special case of Zhao and Ermon [2021], and stakes may be restricted to $b_t \in [0, M]$ without weakening the deployer’s guarantee. This restriction makes the mechanism an *insurance* scheme rather than a two-sided betting market, since deployers only ever purchase coverage against the harm, never a position that bets against the model behaving safely. With $b_t \geq 0$, the side payment is equivalent to paying an unconditional premium of $b_t(\mu_t + c_t)$ in exchange for a payment of b_t if the incident occurs, so the per-unit premium rate $\mu_t + c_t$ decomposes into the forecast and an *ambiguity loading* in the sense of Kunreuther et al. [1993], or an extra premium paid on top of the expected value of loss.

On the lab’s side, the object of interest is the loss from betting, written as

$$\frac{1}{T} \sum_t [b_t(y_t - \mu_t) - b_t c_t].$$

A forecast sequence is *asymptotically exact* if this average loss converges to zero.

Theorem 1 (Zhao and Ermon, 2021, Thm. 1). *For bounded stakes $b_t \in [0, M]$, there exists an online algorithm whose forecasts are asymptotically exact against every strategy of Nature and of the agents, with squared average loss $O(\sqrt{\log T}/\sqrt{T})$.*

In each period, the lab collects the loading $b_t c_t$, which represents the premium loading charged on top of the point forecast. The lab then pays out $b_t(y_t - \mu_t)$, the claims generated by whatever its forecast got wrong. By Theorem 1, there is a way to adjust the posted c_t over time so that, no matter how Nature picks outcomes and no matter how strategically the deployers choose their stakes, these two lines cancel as the margin collected converges to claims paid, and the insurance book nets to zero. If payouts are outpacing margin, past loadings were too thin, so c_t should increase; if margin has outpaced payouts, loadings were too fat, so c_t needs to be cut.

The endpoint of the adjustments is the point where both streams settle on the lab’s realized mispricing on the bets it faced. Payouts sit there by definition, since $b_t(y_t - \mu_t)$ is the cost of forecast error. Margin, however, is driven to that quantity through the algorithm, since any gap between the two in either direction is a cumulative win or loss that the algorithm responds to by moving c_t . Thus, c_t settles at what we will refer to as the *stationary loading*, or the lab’s average forecast error per unit of coverage, calculated over the actual instances it faced. The betting mechanism then results in non-positive loss for the lab in the long run.

3.4 Evaluation Effort

In Zhao and Ermon [2021], generating the forecast is zero-cost. For AI labs, the forecast is the output of a safety evaluation, which requires investment into red-teaming, adversarial prompt elicitation, coverage of languages and personas, and more. Given this, we model the lab as choosing an *evaluation effort* $e \in \mathbb{R}_{\geq 0}$ once, at $t = 0$, at cost κe^2 with $\kappa > 0$. The quadratic reflects how the first adversarial prompt families catch the obvious failures cheaply, and each further increment hunts rarer, better-hidden failure modes. In the closest measured analogue, software security testing, discovering additional distinct failures within a fixed budget requires exponentially growing resources [Böhme and Falk, 2020].

More effort dedicated to evaluating the safety of the model does not change the true level of risk associated with it, or μ_t^* . Instead, dedicating more effort for evaluations sharpens the lab’s

forecast of its model’s risk. This mirrors the evaluation literature’s own architecture. For instance, testing effort determines the bias of the *estimator* of a latent danger level while the danger level itself evolves exogenously [Bova et al., 2024].

The stationary loading of the online adjustment is the forecaster’s realized error, and this realized error is a property of how precise the forecaster’s initial estimates were. This is determined before any deployment by how much *effort* was expended to uncover the context-driven variation in μ_t^* during evaluation. Evaluation effort e is thus how a lab would reduce the stationary loading, and we summarize its effect as a map:

Assumption 1 (Evaluation technology). *At effort e , the online algorithm’s posted loading c_t admits a stationary level $\beta(e)$, which represents the lab’s long-run coverage-weighted forecast error on non-repeating deployment contexts. We set $c_t = \beta(e)$ for all t , where $\beta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{> 0}$ is twice differentiable, strictly decreasing, and convex, with $\lim_{e \rightarrow \infty} \beta(e) = \underline{\beta} > 0$.*

Assumption 1 makes two simplifications for tractability. First, instead of modeling the entire adjustment path of c_t , we model the path’s destination of stationary loading. In reality, c_t takes some number of engagements to settle at the stationary loading, and during that time, deployers face premium rates that differ from $\mu_t + \beta(e)$. By Theorem 1, this discrepancy shrinks at a rate $O(\sqrt{\log T/T})$, so it vanishes from any quantity averaged over the horizon. Second, the assumption takes for granted that the loading has a resting level at all. Strictly, Theorem 1 guarantees only that margin and claims balance on average, so it remains possible that a loading keeps oscillating around the right average. Assuming a stable level can be justified empirically through the simulations of Zhao and Ermon [2021], which show that posted loadings settle quickly and large values are rare (Fig. 3). These simplifications center the paper on long-horizon analysis of the insurance scheme, not short-term deployment horizons that depend more on the adjustment path itself.

The strictly positive floor $\underline{\beta}$ represents safety risks that are structurally impossible to evaluate pre-deployment, such as post-deployment elicitation of dangerous behavior.

3.5 Lab Participation and Insurance Fee

Costly evaluation breaks the participation guarantee of the inherited framework. Under asymptotic exactness the betting book nets to zero, so the lab’s payoff from operating the scheme is $-\kappa e^2 < 0$ for any $e > 0$. Thus, the nonpositive-loss guarantee that suffices for forecaster participation in Zhao and Ermon [2021] no longer does. If the scheme is to operate, someone must pay for it, and the natural payer is the party it creates surplus for. Deployers gain from the scheme precisely what the forecaster’s guarantee is worth to them. We now derive that value and let the lab charge for access to it in order to restore participation.

To examine the deployer’s surplus, we note that deployers cannot verify the lab’s forecast. Given this, we take the cautious (maxmin) benchmark of Zhao and Ermon [2021] and assume deployer t evaluates actions under the worst-case probability $\tilde{\mu} \in [0, 1]$. In the relevant cases where $D_t > v_t$ (potential harm from deploying exceeds deployment value, the configuration that motivates insurance), the worst case of deploying results in a loss, so she abstains and receives 0.

Inside the scheme, Proposition 1 characterizes her optimal payoff:

Proposition 2 (Deployer guarantee and surplus). *Suppose deployer t participates, deploys ($a_t = 1$), and selects coverage $b_t = D_t$. Then her expected payoff from the insurance mechanism equals $u_t = v_t - D_t(\mu_t + \beta(e))$ under any realization of μ_t^* . Her optimal participation payoff is therefore $S_t := (v_t - D_t(\mu_t + \beta(e)))_+$, attained by deploying with full coverage when $v_t > D_t(\mu_t + \beta(e))$ and abstaining otherwise.*

We prove that S_t is a unique optimal payoff for the deployer in Section 4.1. Thus, the deployer’s willingness to pay for model access is thus exactly S_t , and it is strictly decreasing in the loading. Sharper evaluations from the lab makes coverage cheaper to purchase by decreasing the ambiguity loading, which both increases the surplus of each deployer and expands the set of deployers willing to insure. Given this deployer surplus, we restore lab participation by allowing labs to charge a *participation fee* at the time of contracting.

Definition 1 (Participation fee). *Given deployer surplus S_t , the lab charges each participating deployer a participation fee*

$$\varphi_t = \alpha S_t, \quad \alpha \in (0, 1],$$

where α parameterizes the lab’s market power.

This fee is the equivalent of the insurance component of a licensing agreement or a subscription charge for insurance. In this expression, α represents the lab’s market power, not in terms of its market share, but its pricing power and ability to capture the surplus its product creates. The value of α captures several factors, such as how substitutable the lab’s model is and deployers’ switching costs. In a scenario with one monopolistic lab, the take-it-or-leave-it dynamics may mean $\alpha = 1$, while competition between firms or bargaining by deployers may lower α . Moreover, the participation fee makes it so that both the lab and deployer have incentives to opt into the scheme. The deployer retains $(1 - \alpha)S_t \geq 0$, strictly positive when $\alpha < 1$, and we will go on to prove the fee restores lab participation in Section 4.

3.6 Protocol

Protocol 2’ (Toxicity Insurance with Costly Evaluation).

At $t = 0$:

0. The lab chooses evaluation effort $e \geq 0$ (the breadth of adversarial prompt families probed, person-hours of red-teaming, or coverage of languages and personas), incurring one-time cost κe^2 , $\kappa > 0$. This fixes the loading $c_t = \beta(e)$ for the horizon (Assumption 1).

For each deployment engagement $t = 1, \dots, T$:

1. Nature reveals the deployment context $x_t \in \mathcal{X}$ to the lab and chooses $\mu_t^* \in [0, 1]$ without revealing it.
2. The lab performs an evaluation at effort e and announces μ_t and $c_t = \beta(e)$, with $(\mu_t - c_t, \mu_t + c_t) \subset (0, 1)$, posting the insurance schedule: any coverage $b \in [0, M]$ at premium rate $\mu_t + c_t$ per unit.
3. Deployer t , knowing her private value and incident damage, (v_t, D_t) and observing (μ_t, c_t) , decides whether to participate; if so she pays the fee φ_t , chooses $a_t \in \{0, 1\}$, and selects coverage $b_t \in [0, M]$.
4. Nature samples $y_t \sim \text{Bernoulli}(\mu_t^*)$ and reveals it.
5. The deployer’s total loss is $l_t(a_t, y_t) - b_t(y_t - \mu_t) + b_t c_t + \varphi_t$; the lab’s loss on the engagement is $b_t(y_t - \mu_t) - b_t c_t - \varphi_t$.

Lab's **total loss** over the horizon:

$$L_T^{\text{total}} = \sum_{t=1}^T \left[\underbrace{b_t(y_t - \mu_t)}_{\text{coverage payouts}} - \underbrace{b_t c_t}_{\text{ambiguity loading}} - \underbrace{\phi_t}_{\text{fee revenue}} \right] + \underbrace{\kappa e^2}_{\text{evaluation cost}}.$$

4 Results

4.1 The Deployer's Guarantee

We first prove Proposition 2 to establish that full coverage guarantees a payoff of $v_t - D_t(\mu_t + \beta(e))$ regardless of the true probability, and S_t is the deployer's optimal payoff in comparison to all other stakes or abstaining from the insurance mechanism.

Proof. We fix an engagement t and drop the subscript t for the remainder of the proof for concision, and write $c = \beta(e)$. A deployer choosing action $a[tq : \in \{0, 1\}]$ and an arbitrary stake $b \in [0, M]$ receives realized payoff

$$U(a, b, y) = -l(a, y) + b(y - \mu) - bc.$$

First, deploying with full coverage $b_t = D_t$ yields a deterministic payoff. Consider the particular stake $b = D$ (feasible since $D \leq M$) with $a = 1$:

$$U(1, D, y) = v - Dy + D(y - \mu) - Dc = v - D(\mu + c) \quad \text{for every } y \in \{0, 1\},$$

where the $-Dy$ from the loss and the $+Dy$ from the side payment cancel, leaving no dependence on the realized outcome. The payoff is therefore deterministic, and it equals $v - D(\mu + c)$ in expectation under any μ^* , which is the equality case of Proposition 1 for the loss (1).

Second, full coverage is optimal given deployment. The deployer evaluates uncertain payoffs at the worst case $\tilde{\mu} \in [0, 1]$. Thus,

$$\mathbb{E}_{Y \sim \tilde{\mu}} U(1, b, Y) = v - D\tilde{\mu} + b(\tilde{\mu} - \mu) - bc = v - b(\mu + c) - (D - b)\tilde{\mu}.$$

If $b < D$ the coefficient on $\tilde{\mu}$ is negative, so the worst case is $\tilde{\mu} = 1$, giving $v - D + b(1 - \mu - c) < v - D + D(1 - \mu - c) = v - D(\mu + c)$, where the inequality uses $1 - \mu - c > 0$. If $b > D$ the worst case is $\tilde{\mu} = 0$, giving $v - b(\mu + c) < v - D(\mu + c)$, using $\mu + c > 0$. Since b was arbitrary, $b = D$ uniquely maximizes the worst-case payoff of deploying, with value $v - D(\mu + c)$.

Third, participation is worthwhile exactly when that payoff is nonnegative. Under abstention, $\mathbb{E}_{Y \sim \tilde{\mu}} U(0, b, Y) = b(\tilde{\mu} - \mu) - bc$, whose worst case ($\tilde{\mu} = 0$) equals $-b(\mu + c) < 0$ for any $b > 0$; so the abstaining deployer optimally takes $b = 0$ and receives 0. Comparing the participation to abstention, she deploys precisely when $v - D(\mu + c) \geq 0$ (with indifference resolved toward deploying at equality), and her optimal participation payoff is $(v - D(\mu + c))_+ = S_t$. $\square$

This result demonstrates that if the deployer deploys with coverage $b_t = D_t$, her payoff equals $v_t - D_t(\mu_t + \beta(e))$ under any realization of μ_t^* . Moreover, that payoff, floored at zero, is the best she can achieve from the scheme. Establishing that S_t is her optimal surplus, not merely one available payoff, is what allows the fee in Section 3.5 to exist. The fee prices the deployer's best use of the scheme, which is now well-defined.

4.2 Optimal Evaluation Effort

Given the deployer's surplus, we now consider the lab's insurance and evaluation decisions at $t = 0$. The protocol's total loss (3.6) contains three components: betting transfers, fee revenue, and the evaluation cost. Proposition 2 pins down the first two at optimal deployer behavior. For each deployer t , we can write her surplus at loading c as

$$S_t(c) := (v_t - D_t(\mu_t + c))_+.$$

Given this, each deployer participates and pays $\varphi_t = \alpha S_t(\beta(e))$ exactly when that surplus is positive, so fee revenue is $\alpha \sum_t S_t(\beta(e))$ for the lab, while by Theorem 1 the betting component vanishes in the long run. Writing in terms of the lab's profit instead of its loss, the lab's payoff as a function of its one choice variable is

$$\Pi(e; T) = \alpha \sum_{t=1}^T S_t(\beta(e)) - \kappa e^2, \quad (2)$$

where each $S_t(\cdot)$ is continuous, nonincreasing, and convex, bounded by M [more explanation?]. We characterize the lab's optimal evaluation effort in two ways. First, we make no assumptions whatsoever about the sequence of deployers and demonstrate that optimal effort exists and increases with deployment scale. Second, in order to show that effort is nonzero and further characterize where it settles, we place some restrictions on what demand may look like.

Theorem 2 (Existence and scaling of evaluation effort). *For any sequence of deployer types bounded by M :*

- (i) **Existence.** *For every $T \geq 1$, a maximizer of the lab's profit (2) exists; let $e^*(T)$ denote the largest maximizer.*
- (ii) **Effort rises with deployment scale.** *$e^*(T)$ is nondecreasing in T .*
- (iii) **Sublinear growth.** *$e^*(T) \leq \sqrt{\alpha TM/\kappa}$, so $e^*(T) = O(\sqrt{T})$.*

Proof. (i) Given a fixed T , each map $e \mapsto S_t(\beta(e))$ is a composition of continuous functions (Assumption 1 and S_t , a linear composition of a continuous function $\beta(e)$) and is thus continuous. It follows that $\Pi(\cdot; T)$ is continuous. Moreover, we observe that profit eventually falls in the long run, as revenue is capped at αTM (each engagement's surplus is at most M) while the cost κe^2 grows without bound. Choosing E large enough that $\Pi(e; T) < \Pi(0; T)$ for all $e > E$, no effort beyond E can be optimal, and maximizing over $[0, \infty)$ is equivalent to maximizing over the compact interval $[0, E]$. By the extreme value theorem, the continuous function $\Pi(\cdot; T)$ attains its maximum there, so a maximizer exists. Finally, since maximizers need not be unique absent concavity, we select the largest. The set of maximizers is nonempty, bounded (it lies in $[0, E]$), and closed, so it contains its supremum and the largest maximizer $e^*(T)$ is well defined.

(ii) For $e' > e$, Assumption 1 gives $\beta(e') \leq \beta(e)$, and each S_t is nonincreasing, so $S_t(\beta(e')) - S_t(\beta(e)) \geq 0$ for every t . In other words, raising effort weakly raises every engagement's surplus. For $T' > T$,

$$[\Pi(e'; T') - \Pi(e; T')] - [\Pi(e'; T) - \Pi(e; T)] = \alpha \sum_{t=T+1}^{T'} [S_t(\beta(e')) - S_t(\beta(e))] \geq 0.$$

By Topkis's monotonicity theorem, the largest maximizer $e^*(T)$ is nondecreasing in T .

(iii) Since $e^*(T)$ is optimal and $e = 0$ is feasible, $\Pi(e^*(T); T) \geq \Pi(0; T)$. Writing this out and cancelling the fee terms at $e = 0$,

$$\kappa e^*(T)^2 \leq \alpha \sum_{t=1}^T [S_t(\beta(e^*(T))) - S_t(\beta(0))] \leq \alpha TM,$$

where the second inequality holds because each summand is at most M . Rearranging gives $e^*(T) \leq \sqrt{\alpha TM / \kappa}$. $\square$

Theorem 2 deliberately avoids assuming anything about the sequence of deployers and instead ranks the lab's options against each other. Increasing deployment scale never reduces the amount of evaluation effort chosen by the lab, but under these assumptions we cannot guarantee that effort is invested in the first place. For instance, if every arriving deployer has $v_t = 0$, then $S_t(c) \equiv 0$ at every loading, fee revenue is zero, and $e^*(T) = 0$ for all T . Whether effort has strictly positive value is dependent on the lab's expectation of the market it will serve, so we introduce an assumption that places restrictions on average demand so that it is a stable object the lab can plan for.

Assumption 2 (Demand). *For each uncertainty interval c , average deployer surplus converges: $\frac{1}{T} \sum_{t=1}^T S_t(c) \rightarrow \bar{S}(c)$, and the limit $\bar{S}$ is continuously differentiable.*

Assumption 2 permits any order in which the deployers arrive and arbitrary correlation across deployers. It requires only that, over time, the overall mix of deployers settles into a stable pattern, allowing the lab to plan against average deployer surplus at $t = 0$. Moreover, the smoothness condition on the limit $\bar{S}$ corresponds to deployers' entry prices being spread out rather than concentrated at a single value, so that the market fills gradually as forecasts become more precise.

Both conditions hold, for instance, when deployer types (v_t, D_t, μ_t) are drawn i.i.d. from a population admitting a density. In that case, the convergence follows by the law of large numbers, with $\bar{S}(c) = \mathbb{E}[(v - D(\mu + c))_+]$, and entry prices are spread continuously, making it smooth. This environment (implemented in simulation in Section 5) is sufficient but not necessary, as the assumption itself requires only that the mix of deployers settles.

Under Assumption 2, the lab's problem at $t = 0$ is a choice of effort given expectations of average demand:

$$\max_{e \geq 0} \bar{\Pi}(e; T) = \alpha T \bar{S}(\beta(e)) - \kappa e^2, \quad (3)$$

This revised problem replaces the sum of individual surpluses in (2), which the lab cannot compute at $t = 0$ with no knowledge of the deployer population, with T times the surplus of the average engagement. In the i.i.d. case, $\bar{\Pi}$ is exactly expected profit, and in general, the gap between the long-horizon expected average surplus $\bar{S}(\beta(e))$ and the realized average $\frac{1}{T} \sum_{t \leq T} S_t(\beta(e))$ vanishes by Assumption 2. The per-instance error vanishes as the market grows, so conclusions about the lab's initial planning problem can be applied to the lab's problem at scale. With the $t = 0$ problem now defined, we derive results about how much effort is invested by labs.

Theorem 3 (Positive effort and saturation). *Under Assumptions 1 and 2:*

- (i) **Positive effort.** *If $\bar{S}'(\beta(e)) > 0$, i.e. a marginal unit of effort raises average deployer surplus at zero effort, then $e^*(T) > 0$ for every time horizon $T \geq 1$.*
- (ii) **Saturation at the floor.** *If $\bar{S}'(\beta(e)) > 0$ whenever $\beta(e) > \underline{\beta}$, i.e. evaluation effort retains some marginal value until the floor $\underline{\beta}$ is reached, then $\beta(e^*(T)) \rightarrow \underline{\beta}$ as $T \rightarrow \infty$. Scale drives the loading to its irreducible floor, and no further.*

Proof. (i) By Assumption 2, $\bar{S}$ is continuously differentiable, and β is differentiable by Assumption 1, so $\bar{\Pi}(\cdot; T)$ is continuously differentiable with $\bar{\Pi}'(e; T) = \alpha T \bar{S}'(\beta(e)) \beta'(e) - 2\kappa e$. We observe that at $e = 0$, $\bar{\Pi}'(0^+; T) = \alpha T \bar{S}'(\beta(0)) \beta'(0) > 0$, given the hypothesis of (i) and $\alpha, T > 0$. Thus, $e = 0$ is not a maximizer of (3), and since a largest maximizer exists by Theorem 2(i) applied to $\bar{\Pi}$, it satisfies $e^*(T) > 0$.

(ii) Suppose $\bar{S}'(\beta(e)) \beta'(e) > 0$ whenever $\beta(e) > \underline{\beta}$. If $\beta(0) = \underline{\beta}$, then monotonicity of β gives $\underline{\beta} \leq \beta(e^*(T)) \leq \beta(0) = \underline{\beta}$ for all T , and the conclusion holds trivially. Therefore $\beta(0) > \underline{\beta}$. Then, the hypothesis applies at $e = 0$, yielding $\bar{S}'(\beta(0)) \beta'(0) > 0$, so by Part (i) $e^*(T) > 0$ for every $T \geq 1$ and the first-order condition below is available at every horizon.

We bound the growth of $e^*(T)$, as in Theorem 2(iii). Since $e = 0$ is feasible, optimality gives $\bar{\Pi}(e^*(T); T) \geq \bar{\Pi}(0; T)$. The revenue gain from effort is now capped by $\bar{S}(\underline{\beta}) - \bar{S}(\beta(0))$, i.e. the surplus that evaluation could ever unlock, instead of the less strict bound M . Rearranging,

$$\kappa e^*(T)^2 \leq \alpha T [\bar{S}(\underline{\beta}) - \bar{S}(\beta(0))].$$

Wherever $e^*(T) > 0$, the first order condition holds: $\alpha T \bar{S}'(\beta(e^*(T))) \beta'(e^*(T)) = 2\kappa e^*(T)$. Combining with the bound above, we have

$$0 \leq \bar{S}'(\beta(e^*(T))) \beta'(e^*(T)) = \frac{2\kappa e^*(T)}{\alpha T} \leq 2\sqrt{\frac{\kappa [\bar{S}(\underline{\beta}) - \bar{S}(\beta(0))]}{\alpha T}} \rightarrow 0,$$

with the leftmost inequality holding since $\bar{S}' \leq 0$ and $\beta' \leq 0$.

Since $e^*(T)$ is nondecreasing, it converges to some $\bar{e} \in (0, \infty]$. There are two cases for $\bar{e}$:

- (1) If $\bar{e} = \infty$, then $\beta(e^*(T)) \rightarrow \underline{\beta}$ directly from $\lim_{e \rightarrow \infty} \beta(e) = \underline{\beta}$ (Assumption 1).
- (2) If $\bar{e} < \infty$, then the map $e \mapsto \bar{S}'(\beta(e)) \beta'(e)$ is continuous (a product of compositions of continuous functions), so its value at $\bar{e}$ is the limit of its values along $e^*(T)$, namely zero; the hypothesis of (ii) states the product is strictly positive wherever $\beta > \underline{\beta}$, so $\beta(\bar{e}) = \underline{\beta}$, and continuity of β gives $\beta(e^*(T)) \rightarrow \beta(\bar{e}) = \underline{\beta}$.

In either case $\beta(e^*(T)) \rightarrow \underline{\beta}$. □

Theorem 3 builds on the results of Theorem 2(ii). We first establish that evaluation effort increases with deployment scale because the costs of that effort are fixed and amortized over deployment instances, while the benefits of decreasing the stationary loading of every engagement scales with the number of deployers. In Theorem 3, we show that in a market of deployers that care about the precision of evaluation, labs invest a strictly positive amount of effort which grows with deployment scale until the uncertainty reaches the irreducible floor $\underline{\beta}$. Evaluation, when well-funded, sharpens evaluations but cannot fully resolve post-deployment uncertainty.

4.3 Lab Participation

Introducing costs of evaluation raised the question of whether the lab operates the scheme at all. Under exactness the betting book nets to zero, so absent fees the lab's payoff would be $-\kappa e^2$. We observe that given the lab's effort is chosen optimally, the lab never sinks an evaluation cost the anticipated fees do not cover. Let the lab's outside option be not operating the scheme, which under the maxmin benchmark of Section 3.5 earns nothing from cautious deployers, and normalize it to zero.

We first observe that labs weakly participate unconditionally. Under Assumption 2, for every $T \geq 1$ and every $\alpha \in (0, 1]$, the lab's optimized profit satisfies $\bar{\Pi}^* := \max_{e \geq 0} \bar{\Pi}(e; T) \geq 0$: $e = 0$

is always feasible for labs, and $\bar{\Pi}(0; T) = \alpha T \bar{S}(\beta(0)) \geq 0$, since $\bar{S} \geq 0$ as a limit of averages of nonnegative terms. Moreover, if $\bar{S}(\beta(0)) > 0$, i.e. some deployers gain surplus from the scheme even at zero evaluation effort, then participation is strict, as $\bar{\Pi}^* \geq \bar{\Pi}(0; T) > 0$ for every $T \geq 1$.

Suppose instead that $\bar{S}(\beta(0)) = 0$, i.e. deployers do not gain surplus at zero evaluation effort, and the lab must invest in evaluation to create the market. For each choice of effort $e > 0$, the lab incurs a cost of κe^2 while expecting a gain of $\alpha \bar{S}(\beta(e)) T$. We will call an effort level $e > 0$ *demand-unlocking* if $\bar{S}(\beta(e)) > 0$, and for each such e we will call $\kappa e^2 / (\alpha \bar{S}(\beta(e)))$ its *break-even horizon*, i.e. the deployment scale where gains begin to outweigh costs. Then,

$$\hat{T} := \inf \left\{ \frac{\kappa e^2}{\alpha \bar{S}(\beta(e))} : e > 0, \bar{S}(\beta(e)) > 0 \right\},$$

is the smallest break-even horizon where effort unlocks demand. If no effort level is able to create demand, the set is empty and $\hat{T} = \infty$ by the usual convention, so no horizon clears the threshold and the market cannot exist.

Proposition 3 (Market emergence). *Under Assumption 2, suppose $\bar{S}(\beta(0)) = 0$. Then participation is strict, $\bar{\Pi}^* > 0$, if and only if $T > \hat{T}$. The operating lab then chooses strictly positive evaluation effort and participates.*

Proof. Suppose first that $T > \hat{T}$. Since T exceeds the infimum of the break-even horizons, some demand-unlocking effort's break-even horizon lies strictly below T . There is $e_0 > 0$ with $\bar{S}(\beta(e_0)) > 0$ and $\kappa e_0^2 / (\alpha \bar{S}(\beta(e_0))) < T$. Rearranging, $\alpha T \bar{S}(\beta(e_0)) > \kappa e_0^2$, i.e. $\bar{\Pi}(e_0; T) > 0$. Hence, the market is large enough to repay e_0 , so $\bar{\Pi}^* \geq \bar{\Pi}(e_0; T) > 0$.

Conversely, suppose $\bar{\Pi}^* > 0$. Since investing in no effort earns exactly zero profit here, any maximizer e^* satisfies $e^* > 0$. Writing out $\bar{\Pi}(e^*; T) > 0$ gives $\alpha T \bar{S}(\beta(e^*)) > \kappa (e^*)^2 \geq 0$, which requires $\bar{S}(\beta(e^*)) > 0$. Rearranging the same inequality, its break-even horizon satisfies $\kappa (e^*)^2 / (\alpha \bar{S}(\beta(e^*))) < T$. Since $\hat{T}$ is the infimum over all demand-unlocking efforts and e^* is one of them, $\hat{T}$ is at most this ratio, so $\hat{T} > T$.

This proves the equivalence. The converse direction also establishes the final claim, since whenever the lab operates at strict profit in this case, every maximizer is strictly positive. Hence, a profitably operating lab necessarily evaluates. $\square$

This proposition demonstrates that the participation fee structure is able to restore participation from labs, despite the costs of evaluation effort. We go further to examine how levels of uncertainty affect whether labs opt into the scheme. When the forecast at $e = 0$ is too uncertain for deployers to gain a surplus, the scheme is profitable only above a minimum deployment scale $\hat{T}$, and the lab necessarily evaluates in order to operate.

4.4 Market Power and Evaluation Incentive

The lab's reward for producing sharper evaluations is the share α of deployer surplus it can claim, which raises the question of how the market structure surrounding the scheme affects the evaluation effort inside it.

Corollary 1 (Market power and the evaluation incentive). *Making no assumptions about the sequence of deployers, Π has increasing differences in (e, α) , so the largest maximizer e^* is nondecreasing in α .*

Proof. For $e' > e$ and $\alpha' > \alpha$,

$$[\Pi(e'; \alpha') - \Pi(e; \alpha')] - [\Pi(e'; \alpha) - \Pi(e; \alpha)] = (\alpha' - \alpha) \sum_{t \leq T} [S_t(\beta(e')) - S_t(\beta(e))] \geq 0,$$

term by term as in Theorem 2(ii), so Π has increasing differences in (e, α) and the largest maximizer e^* is nondecreasing in α by the same argument. The computation applies in the same way to expected profit $\bar{\Pi}$ under Assumption 2. $\square$

Market power is therefore beneficial for how the scheme incentivizes evaluation effort. A larger α allows an individual lab to capture more of the surplus of deployers, in turn making it more attractive for labs to increase total deployer surplus by sharpening evaluations. However, market power can also be harmful to deployers, as bargaining from deployers or competition between labs makes each individual transaction more favorable, as the deployer retains $(1 - \alpha)S_t$ of her surplus. We return to the implications of market structure in Section 6.

5 Numerical Analysis

We use simulation to verify that the inherited mechanism behaves as required in our deployment setting. Empirical calibration is not available for either exercise, which follows from the motivation of the paper. No AI harm insurance market currently operates, so there is no premium or claims data to fit; incident data for deployed models is sparse, non-standardized, and largely non-public; and the true incident probability μ_t^* is unobservable, since each deployment context differs in risk.

We implement the online mechanism of Zhao and Ermon [2021]. We illustrate the two behaviors the paper inherits from the mechanism: first, that the lab’s average betting loss vanishes even with flawed forecasting, and secondly, that the loading settles at the level of realized error.

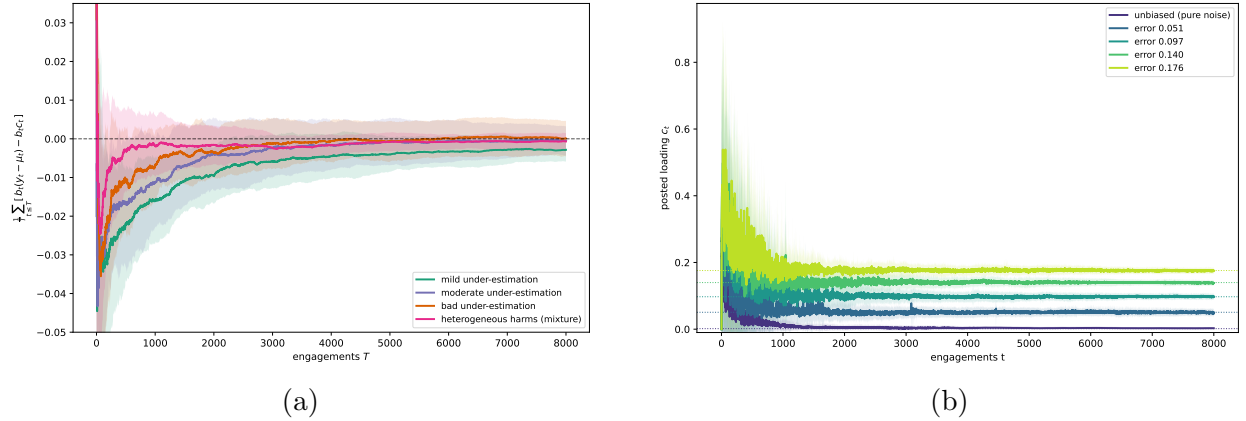


Figure 1: The lab’s insurance book. (a) Average betting loss across deployment engagements converges to zero under biased base forecasts of increasing severity and under heterogeneous harms (Theorem 1). (b) The posted loading c_t over the horizon settles at a stationary loading that is the error of the base predictor, demonstrating Assumption 1.

Figure 1(a) tests whether the lab’s loss converges to zero in the situations in which safety evaluations systematically under-estimate risk at three severities, and in a fourth environment deployers draw from a mixture of heterogeneous harm processes. These are the situations where naive insurance is problematic, as premiums are collected as if risks are rare while claims arrive at

the higher true rate. We show that the online correction is able to resolve this problem by initially posting large c_t values and then adjusting to create non-negative loss in all cases.

Figure 1(b) illustrates Assumption 1. Holding the environment fixed, we vary the safety evaluation’s overall error, $|\mu_t^* - \mu_t|$, and graph the posted loading c_t across engagements. First, we observe the loading settles after an initial adjustment and each path converges towards a stable level, which is the behavior Assumption 1 summarizes when it replaces the adjustment path with a stationary loading. Second, the stationary loading are ordered by forecast quality, as increasing initial error leads to increasing stationary loading. This empirically demonstrates the map $\beta(e)$, as effort into more accurate evaluations lowers the stationary loading.

Taken together, these results support Theorem 1 and Assumption 1 as the foundation of the long-run analysis.

6 Discussion

Taken together, our results describe a mechanism that governs the risks of AI deployment by pricing that risk rather than specifying regulatory guidelines *ex ante* or litigating harms *ex post*. The mechanism is able to do so despite the unreliability of safety evaluations, because insurance loadings correct for incorrect forecasts and weakly benefit all deployers (Proposition 2). Insurance also funds the improvement of the very information it runs on: sharper evaluations allow for higher fees, so a lab serving a sufficient number of deployers invests effort into evaluation without a mandate (Theorem 3), and evaluates more at a greater scale (Theorem 2). Risks beyond the reach of evaluations, moreover, are visible through insurance premiums rather than discovered later on. Overall, the arrangement is weakly Pareto-improving relative to the no-scheme benchmark: participating deployers retain $(1 - \alpha)S_t \geq 0$ and the lab profits above the threshold scale.

The incentive for evaluation effort is financed by the captured deployer’s surplus, and in turn, our results suggest market power is complementary in two ways to the evaluation effort incentive. A larger share α raises the lab’s return to sharper evaluations (Corollary 1) and lowers the deployment scale needed for the scheme to be profitable since $\hat{T} \propto 1/\alpha$ (Proposition 3). Frontier AI deployment is currently concentrated among a small number of labs with large engagement volumes and substantial pricing power, which is the large- T and high- α environment in which this mechanism is viable and most strongly incentivizes evaluation effort.

However, high market power also cuts the other way. A lower α , in the form of bargaining from deployers or competition between labs, makes each individual transaction more favorable, as the deployer retains $(1 - \alpha)S_t$ but erodes the incentives for the evaluation effort that could increase surplus overall. Whether the net effect of the per-transaction gain or the evaluation effort loss is ambiguous, and modeling that game is a natural next step from this paper.

Moreover, our model diverges from some existing proposals for insurance by having the lab insure itself, instead of a third-party insurer. We choose this structure due to the importance of the evaluation information. The forecast comes from the lab’s own evaluations, so an outside insurer would have to price coverage using information (such as model weights and evaluation results) that the lab produces and controls. The record of insurance against cyberattacks suggests that insurers cannot independently price system-specific technological risk [Schwarcz and Wolff, 2025]. Making the lab insure itself puts the party that produces the risk estimate at risk for its accuracy.

A separate objection from the governance literature is that self-insurance fails to elicit more care, or efforts to reduce risk, on the part of the lab [Trout, 2025]. However, our model explicitly excludes concerns about mitigating the risks of harm and instead focuses on effort into evaluation, which is encouraged through the mechanism by having labs both generate and insure against their

own forecasts.

Limitations. Several limitations bound our results. First, we assume deployers evaluate unverifiable forecasts at the worst case, the standard maxmin benchmark for ambiguity without a trusted prior [Gilboa and Schmeidler, 1989]. This is most plausible for cautious deployers such as hospitals or government agencies, while milder ambiguity aversion in other deployers would shrink measured surplus. Second, we assume that evaluation effort is only chosen once, with no opportunity for labs to update their effort while running the insurance scheme after learning from realized incidents. Third, AI deployments may carry with them large tail risks [citation?], but our model focuses on harms within an insurable limit M . Outside of that bound, public backstops and capital-market instruments may be necessary, and we defer that analysis [Trout, 2024].

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